ADD ON REVISION COM108

1. Determine if the improper integral $\int_0^5 \ln x\,dx$ convergent. If it converges, compute its value. If it diverges, explain why.

2. ~~Determine the surface area of the solid obtained by rotating $y=\sqrt{9-x^2}$, $-2\le x\le 2$ about the $x-$axis.~~

3. Determine the length of $y=\ln(\sec x)$ between $0\le x\le \frac{\pi}{4}$.

1. $\lim_{t\to 0^+} \int_t^5 \ln x\,dx$

$\int u\,dv = uv - \int v\,du$

let $u = \ln x$, $dv = dx$

$du = \frac{1}{x}dx$, $v = x$

$= \lim_{t\to 0^+} [x\ln x]_t^5 - \lim_{t\to 0^+} \int_t^5 x\cdot\frac{1}{x}dx$

$= \lim_{t\to 0^+} [x\ln x]_t^5 - \lim_{t\to 0^+} [x]_t^5$

$= 5\ln 5 - \lim_{t\to 0^+} t\ln t - 5$

$= 8.047 - \lim_{t\to 0^+} \frac{\ln t}{\frac{1}{t}}$

$= 8.047 - \lim_{t\to 0^+} \frac{\frac{1}{t}}{-\frac{1}{t^2}}$

$= 8.047$

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$\frac{dy}{dx} = \frac{\sec x\tan x}{\sec x} = \tan x$

$\text{length} = \int_0^{\frac{\pi}{4}} \sqrt{1+\tan^2 x}\,dx$

$= \int_0^{\frac{\pi}{4}} \sec x\,dx$

$= \int_0^{\frac{\pi}{4}} \sec x\cdot\frac{\sec x+\tan x}{\sec x+\tan x}\,dx$

$= \int_0^{\frac{\pi}{4}} \frac{\sec^2 x+\sec x\tan x}{\sec x+\tan x}\,dx$

let $u=\sec x+\tan x$; $\frac{du}{dx}=\sec x\tan x+\tan^2 x$ $\Rightarrow dx=\frac{du}{\sec x\tan x+\tan^2 x}$

when $x=\frac{\pi}{4}$, $u=\sec(\frac{\pi}{4})+\tan(\frac{\pi}{4})=\sqrt{2}+1$

$x=0$, $u=\sec(0)+\tan(0)=1$

$= \int_1^{\sqrt{2}+1} \frac{1}{u}\,du = \frac{1}{2}\left[\frac{\ln|u|}{u^2}\right]_1^{\sqrt{2}+1} = \frac{1}{2}(2-0) = 1$

$= \left[\ln|u|\right]_1^{\sqrt{2}+1} = \ln|\sqrt{2}+1| = 0.8814$

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$\lim_{t\to 0^+} \int_t^5 \ln x\,dx$

let $u=\ln x$, $dv=dx$

$du=\frac{1}{x}dx$ $\quad v=x$

$= \lim_{t\to 0^+} [x\ln x]_t^5 - \lim_{t\to 0^+} \int_t^5 x\cdot\frac{1}{x}\,dx$

$= 5\ln 5 - \lim_{t\to 0^+} t\ln t - \lim_{t\to 0^+}[t]_t^5$

$= 5\ln 5 - \lim_{t\to 0^+} \frac{\ln t}{\frac{1}{t}} - 5$

$= 5\ln 5 - \lim_{t\to 0^+} \frac{\frac{1}{t}}{\frac{1}{t^2}} - 5$

$= 5\ln 5 - 5$

1) $\int_0^5 \ln x\,dx = \lim_{t\to 0^+} \int_t^5 \ln x\,dx$ Integration by parts

$= \lim_{t\to 0^+} [x\ln x - x]_t^5$

$= 5\ln 5 - 5 - [\lim_{t\to 0^+} t\ln t - t]$

$= 5\ln 5 - 5$

$\lim_{t\to 0^+} t\ln t = \lim_{t\to 0^+} \frac{\ln t}{\frac{1}{t}}$

$\overset{H}{=} \lim_{t\to 0^+} \frac{\frac{1}{t}}{-\frac{1}{t^2}}$

$= \lim_{t\to 0^+} -t$

$= 0$

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$\frac{dy}{dx} = \frac{\sec x \tan x}{\sec x} = \tan x$

$\text{Length} = \int_0^{\frac{\pi}{4}} \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$

$= \int_0^{\frac{\pi}{4}} \sqrt{1+\tan^2 x}\,dx$

$= \int_0^{\frac{\pi}{4}} \sec x \cdot \frac{\sec x+\tan x}{\sec x+\tan x}$

$= \left[\ln|\sec x+\tan x|\right]_0^{\frac{\pi}{4}}$

2) $\text{Length} = \int_0^{\frac{\pi}{4}} \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$

$= \int_0^{\frac{\pi}{4}} \sqrt{1+\tan^2 x}\,dx$

$= \int_0^{\frac{\pi}{4}} \sec x\,dx$

$= \int_0^{\frac{\pi}{4}} \sec x \times \frac{\sec x+\tan x}{\sec x+\tan x}\,dx$

Let $u = \sec x + \tan x$

$= \left[\ln|\sec x+\tan x|\right]_0^{\frac{\pi}{4}}$

$= \ln(1+\sqrt{2})$

≈ 0.8814

$y = \ln(\sec x)$

$\frac{dy}{dx} = \frac{\sec x \tan x}{\sec x}$

$= \tan x$

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$$dy = \frac{\sec x + \tan x}{\sec x} \, dx$$

$$\frac{dy}{dx} = \frac{\sec x \tan x}{\sec x}$$